Title of your topic

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Abstract

Guidelines for Writing the Abstract:

1. Keep the abstract concise and to the point, providing a brief summary of the main points covered in the report.
2. Ensure that the abstract is clear and easy to understand, avoiding technical jargon or overly complex language.
3. Structure the abstract to include the key elements of the report
4. Ensure that the abstract accurately reflects the content of the report, providing an honest representation of the key points covered.
5. Keep the abstract within a reasonable length, typically around **150-250 words**, to provide a concise overview without going into excessive detail.
6. The abstract should stand alone and not include citations or references to other sources.

Note: The report cover design above is only for reference purpose. Your team may opt for a different report cover if desired.

Acknowledgements

Note: While this section is optional, it is highly recommended that your group completes it.

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Chapter 1

Introduction

1.1 Objective

1.2 Related works

1.2.1 Classical approaches

1.2.2 Deep learning

The introduction of report should contains these contents:

- Overview of your topic and the real-world applications in various fields.
- Briefly introduce the fundamental concepts and applications of your topic.

For example, in the Linear Algebra topic:

- Provide an overview of linear algebra and briefly introduce the importance in various fields such as mathematics, computer science, physics, and engineering.
- Briefly introduce the fundamental concepts and applications of linear algebra.

Chapter 2

Some fundamental concepts of Linear Algebra

Note: This is just a suggestion, and you are free to deviate from the structure provided below, but ensure that all necessary contents are included.

For instance, when preparing the Linear Algebra report, you may adhere to this outline. Feel free to split it into multiple sections if needed.

1. Vector Spaces and Subspaces:

- Define vector spaces and discuss their properties, including closure under addition and scalar multiplication.
- Explain the concept of subspaces and their significance in linear algebra.
- Provide examples of vector spaces and subspaces in different contexts.

2. Linear Independence and Basis:

- Define linear independence and explain its importance in vector spaces.
- Discuss the concept of a basis and its role in spanning a vector space.
- Explain how to determine whether a set of vectors is linearly independent and how to find a basis for a given vector space.

3. Linear Transformations:

- Define linear transformations and discuss their properties, including linearity and preservation of vector operations.
- Explain the matrix representation of linear transformations and the concept of kernel and image.
- Provide examples of linear transformations in geometric and abstract contexts.

Chapter 3

Applications of Linear Algebra

- Provide examples of real-world applications of linear algebra, such as solving systems of linear equations, image processing, data compression, and machine learning.
- Discuss how linear algebra is used in various scientific and engineering disciplines.

Chapter 4

Guidelines for writing the citation

Note: You can choose the style of citation that will be used. However, your citation must be accurate and consistent. Here are the guidelines of using Harvard format:

4.1 Books

Format: Author, A. (Year). *Title of Book* [1]. City, Country: Publisher.

Example: Cormen, T. H., Leiserson, C. E., Rivest, R. L., Stein, C. (2009). *Introduction to Algorithms*. Cambridge, MA: MIT Press.

4.2 Journal Articles

Format: Author, A. (Year). Title of Article. *Title of Journal*, Volume(Issue), Page(s).

Example: LeCun, Y., Bengio, Y., Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436-444.

4.3 Websites

Format: Author/Organization. (Year). Title of Web Page. Available: URL. [Accessed: Month Day, Year].

Example: IEEE. (2023). About IEEE. Available: https://www.ieee.org/about.html. [Accessed: Jan 12, 2023].

4.4 Conference Papers

Format: Author, A. (Year). Title of Paper. In *Proceedings of Conference Name* (pp. xxx-xxx).

Example: Bengio, Y., Courville, A., Vincent, P. (2013). Representation learning: A review and new perspectives. In *Proceedings of the International Conference on Learning Representations (ICLR)* (pp. 1-21).

[1]** means formatting in italic

Bibliography